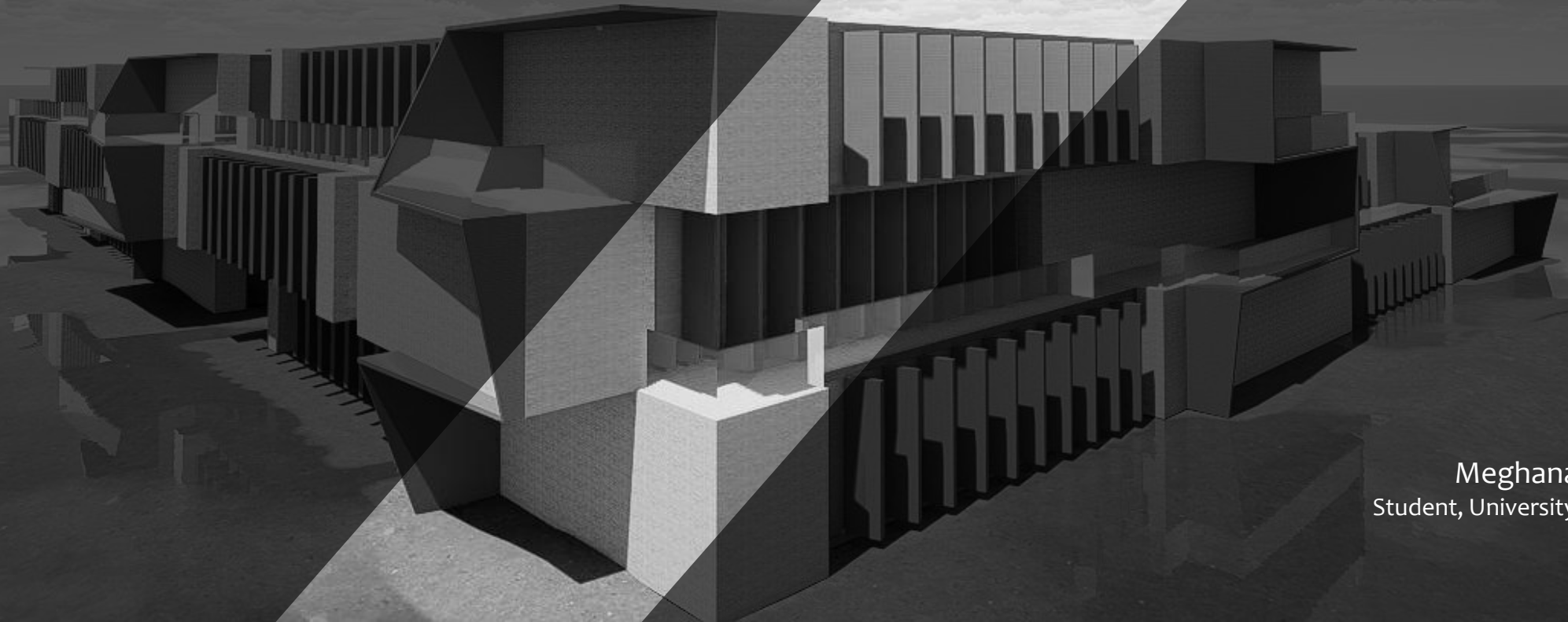


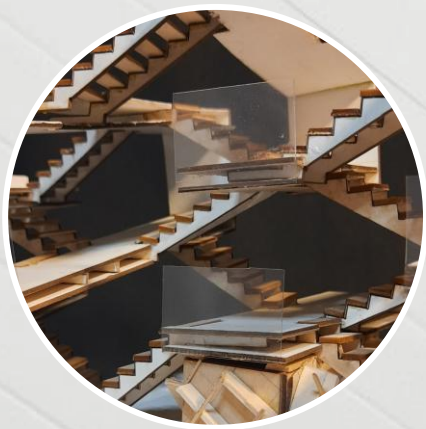
ARCHITECTURE PORTFOLIO



Meghana Kankaria
Student, University of Houston

CONTENT

02



01. The Hive

Educational

05



02. F.I.L.M. Centre

Commercial

09



03. Resource Box

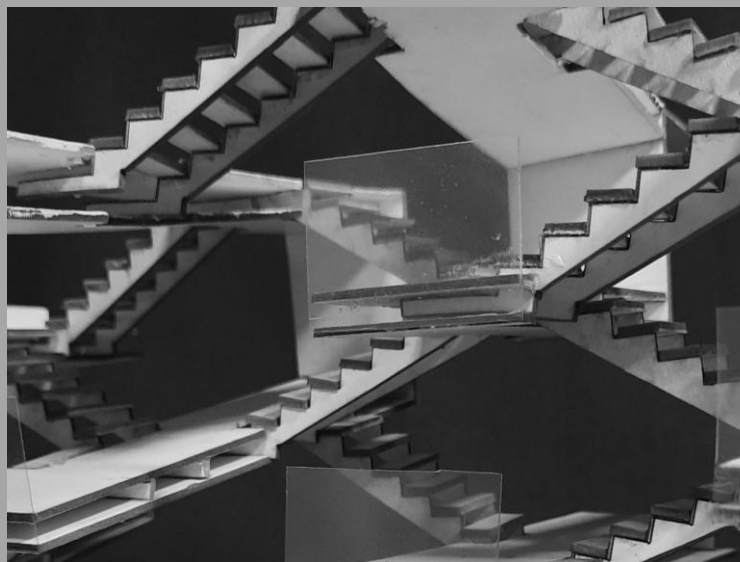
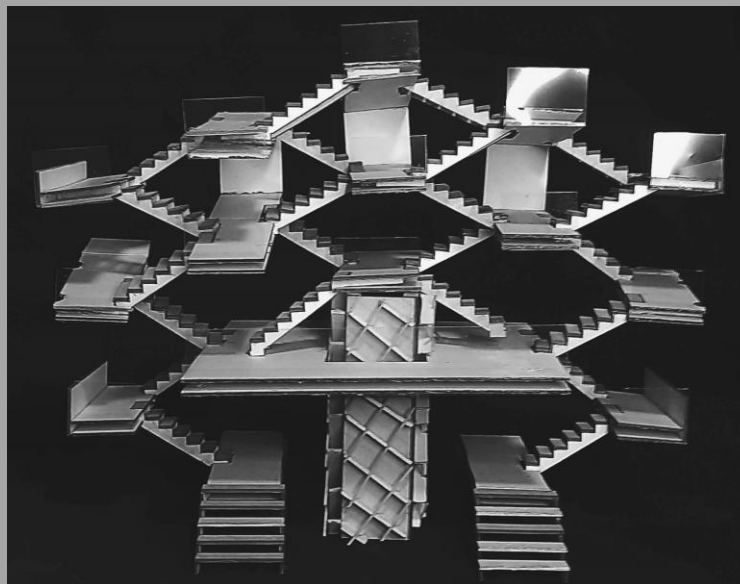
Industrial

13



04. Miscellaneous

Concept

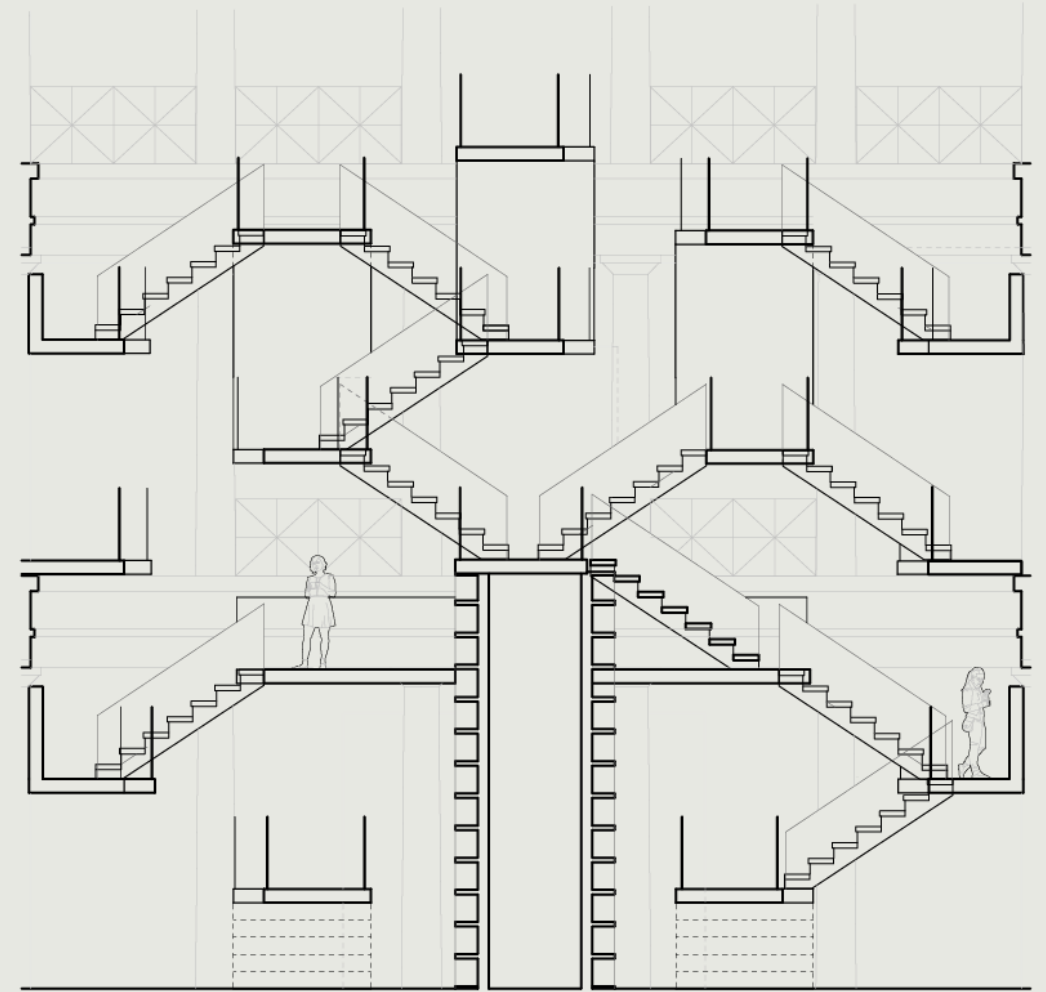
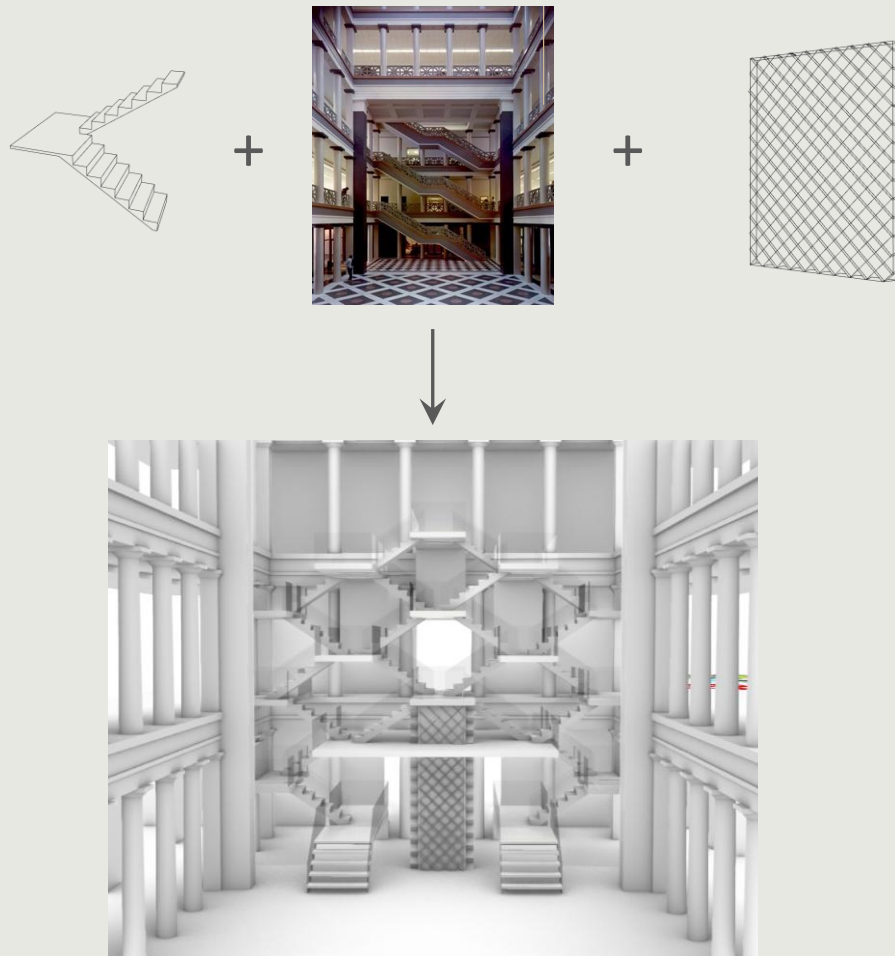


01. THE HIVE

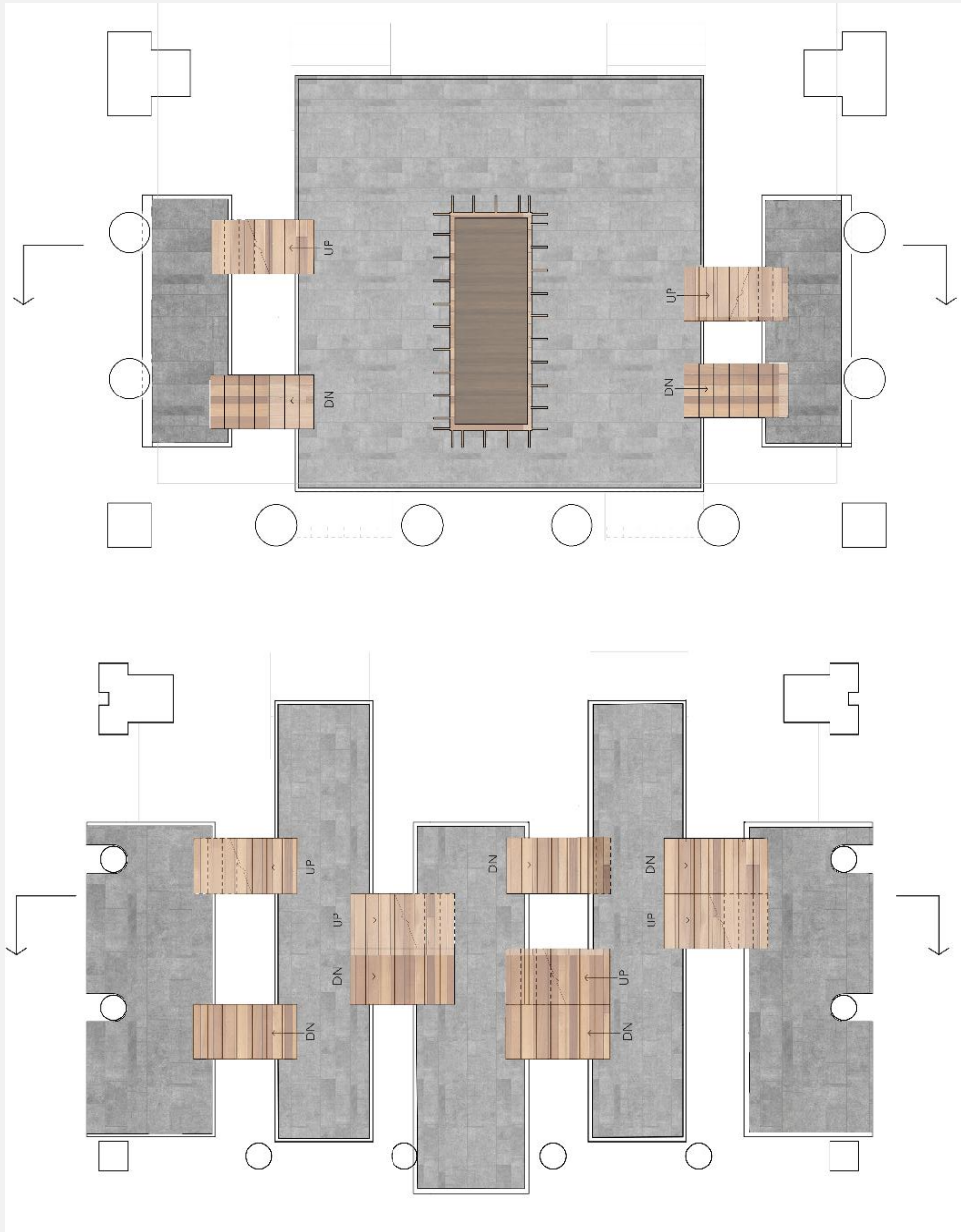
2022

The project was to make an alternate design for the existing set of stairs in the Hines College of Architecture at the University of Houston, which maximizes their functionality, other than being a mode of circulation. **The Hive** draws inspiration from both a beehive and a bustling hub of activity. Designed to complement the atrium's diamond-patterned floor, the honeycomb stair concept features 24 steps across 8 levels, rising to the third floor.

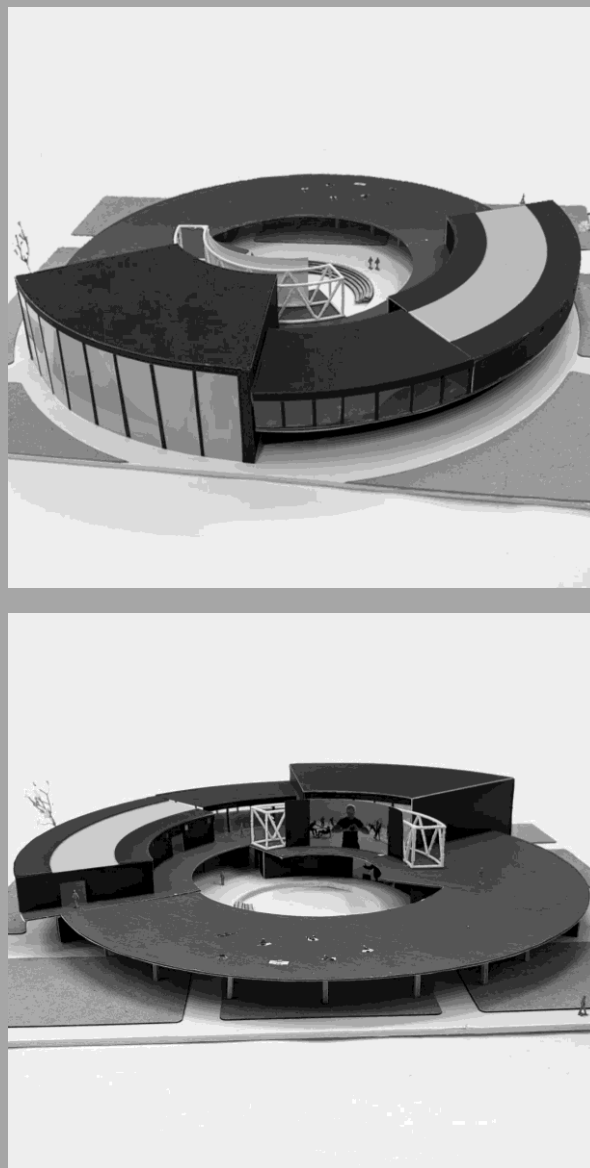
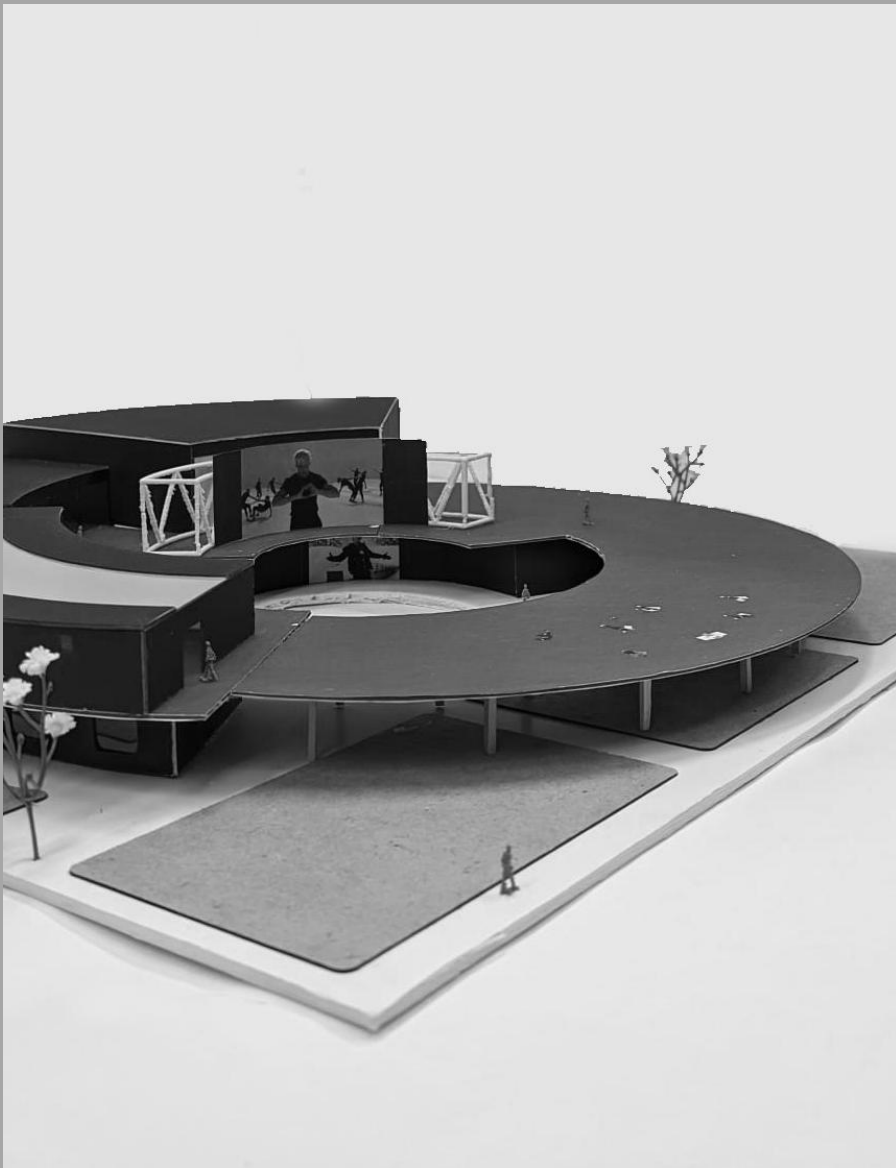
Crafted with wood stairs and bookshelves, glass handrails, and white concrete landings enclosed in glass guardrails, the centerpiece is an 18-foot honeycomb structure displaying books from the CoAD Library. Around it, a 675 sq. ft. platform serves as a viewing deck for atrium events.



The design complements the atrium's diamond-patterned floor while drawing on the strength and stability of honeycomb geometry. This concept not only integrates seamlessly with the architectural language of the space but also allows the landings to remain stable without columns. The result is an open, fluid circulation system that creates both functional movement and a dynamic visual experience within the atrium.



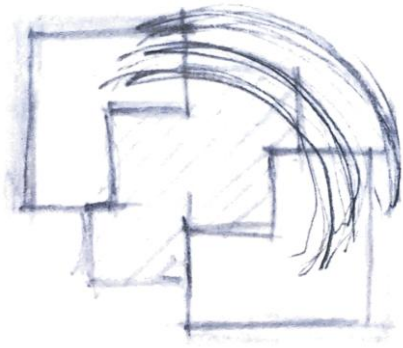
This project provides a dynamic visual journey through the CoAD, highlighting both the library and art gallery. Below the third-floor landing, gallery walls showcase student artwork, while various levels offer spaces to work, gather, and pause. A bookshelf/seating nook at the back invites reading and reflection.



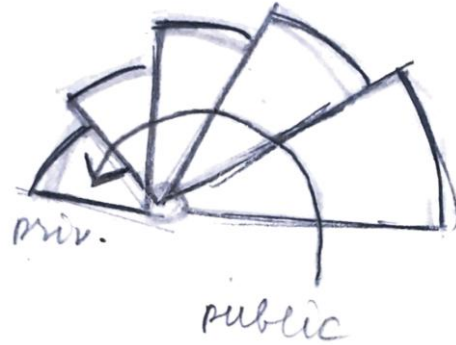
02. F.I.L.M. CENTRE

2023

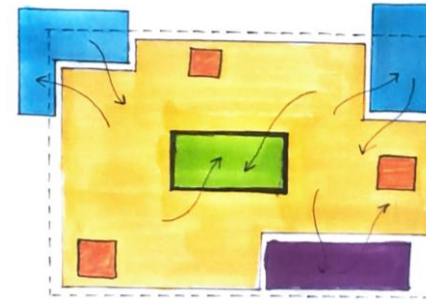
The **F.I.L.M. (Fusion of Immersion, Light, and Motion) Centre** is envisioned as an architectural journey that blurs the line between reality and the cinematic world. Designed as a cultural hub, it provides a space where entertainment, education and storytelling intersect. The circular form embodies continuity and inclusivity, creating an inviting atmosphere for diverse audiences to gather and engage with film in multiple dimensions. It also mimics the form of a cinematic reel. Transparency and solidity are balanced throughout the design, producing rhythms of openness and enclosure that heighten the immersive experience. More than a venue, the F.I.L.M. Building acts as a narrative in itself – an architectural stage where culture, light and motion converge. It invites people to step away from daily life and into a world of creativity, exploration and collective storytelling, reinforcing cinema's enduring power to connect and inspire.



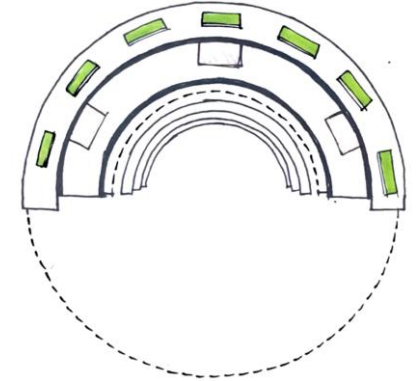
Spatial layering – overlapping forms establish movement and flow within the circular geometry.



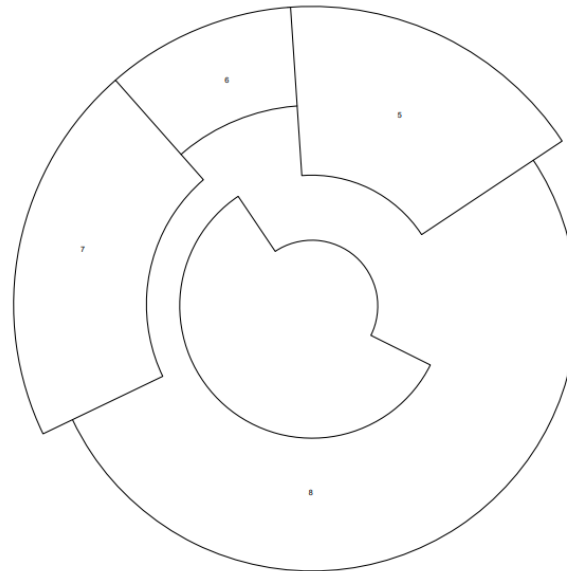
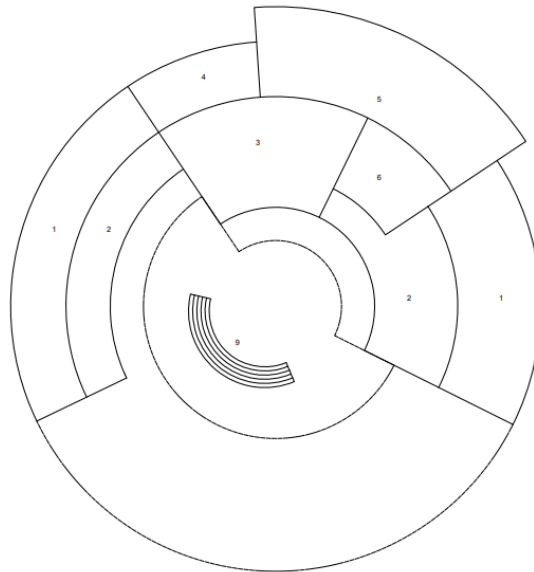
Zoning – Outer layers invite community engagement with amphitheatres and cafés, while the inner core accommodates more intimate and controlled functions such as classrooms and archives.



Program distribution – functional zones arranged around a central core, with circulation weaving connections



Form evolution – amphitheatre seating and program spaces integrated into the circular structure



1. Classrooms/Conference Rooms
2. Library
3. Lobby Area
4. Washrooms
5. Theatre
6. Café
7. Film Gallery
8. Outdoor Theater/Overhang
9. Amphitheatre

The building program includes indoor and outdoor theatres, a film gallery, library, classrooms, a café, workshop and archive spaces and a large amphitheatre. Each element is carefully organized to guide visitors through a transformative sequence—from intimate spaces of learning and reflection to expansive communal zones for performance and celebration.



Solid/Void



Public/Private



Useful/Useless



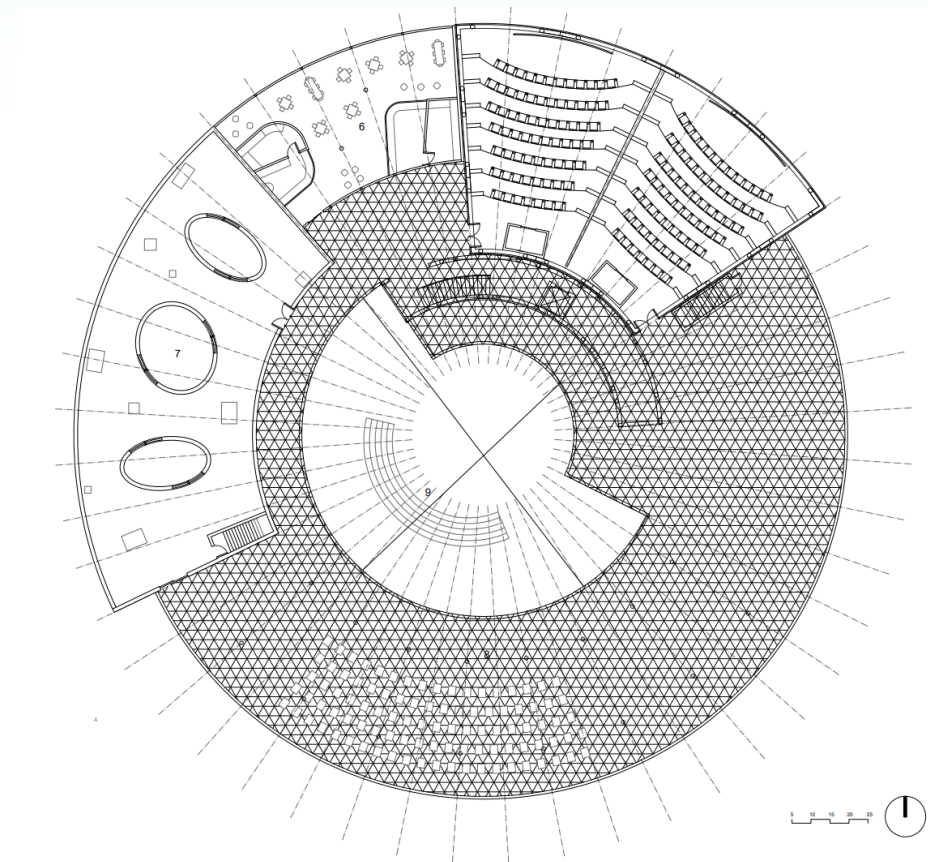
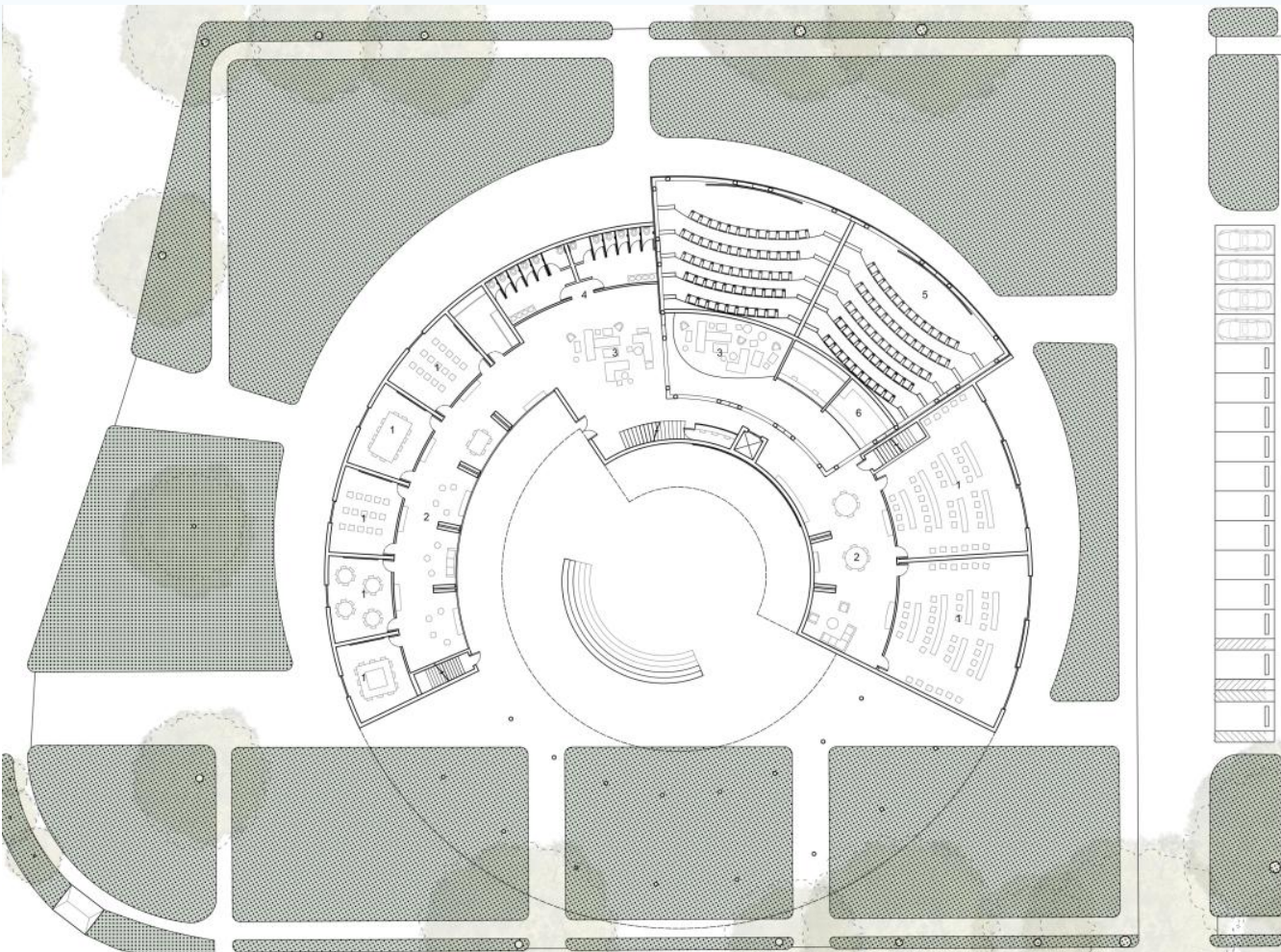
Solid/Void



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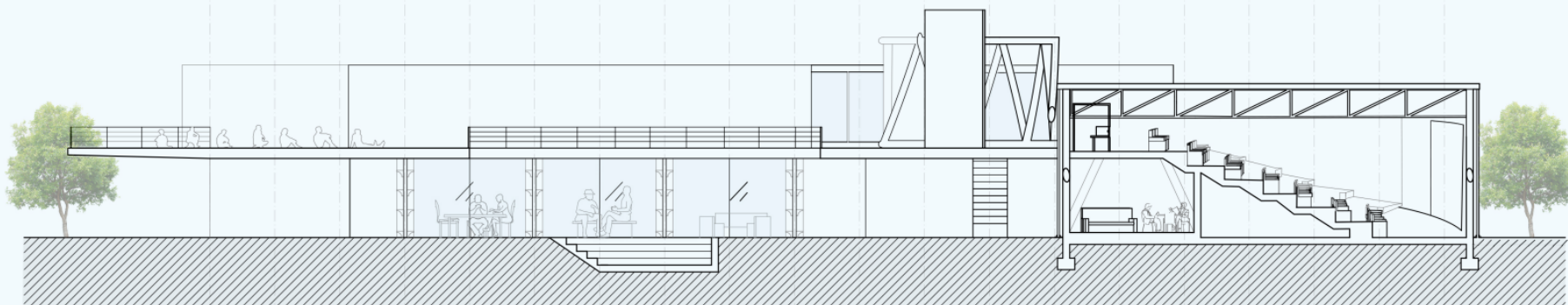


Useful/Useless





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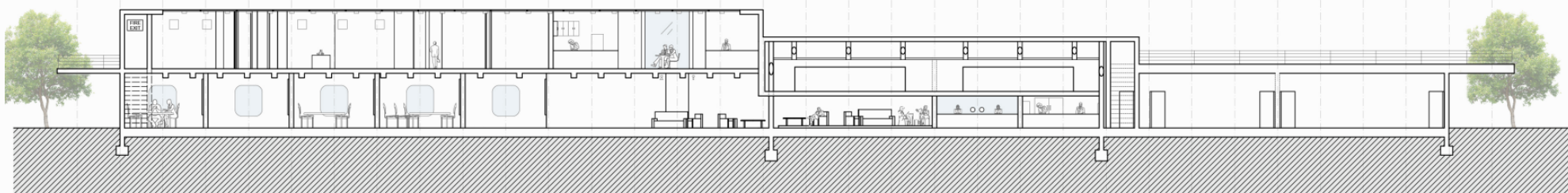
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Materiality



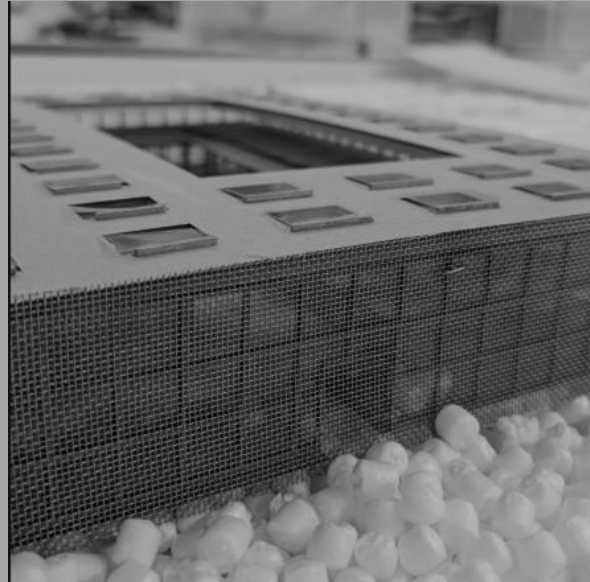
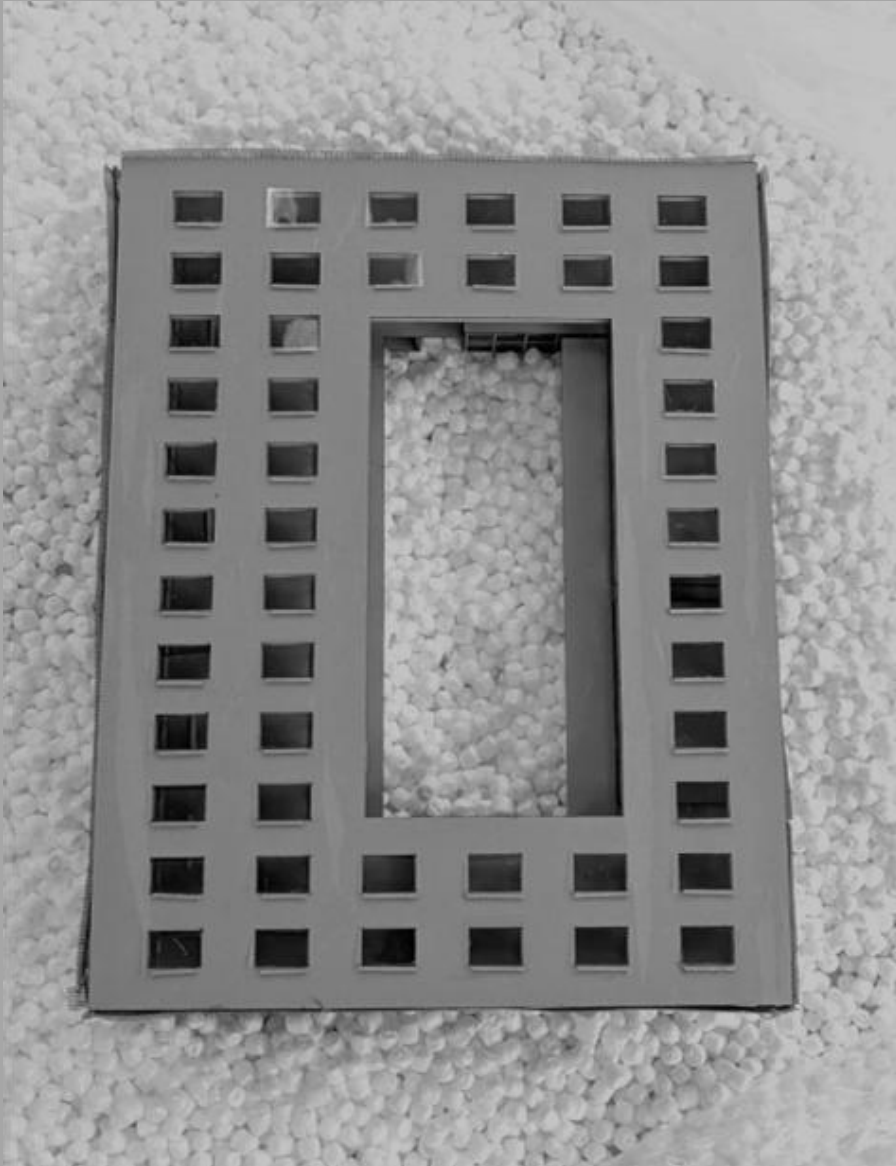
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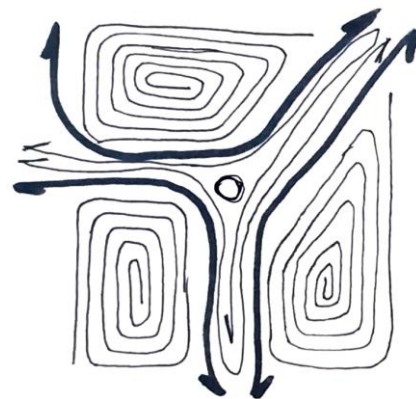
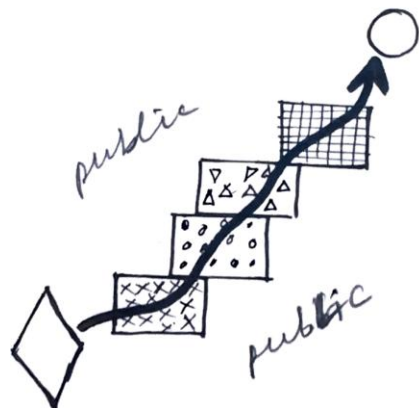
Materiality



03. RESOURCE BOX

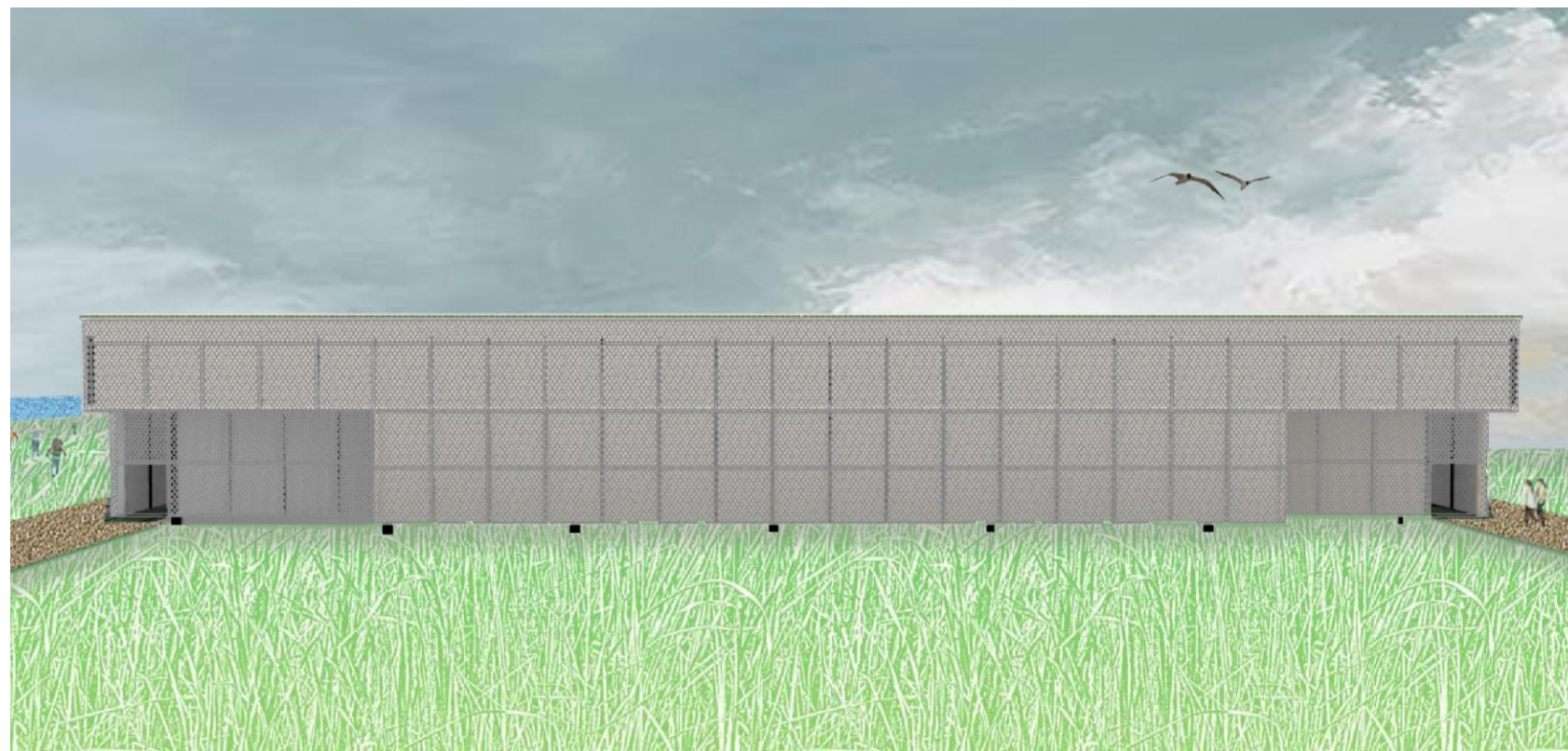
2024

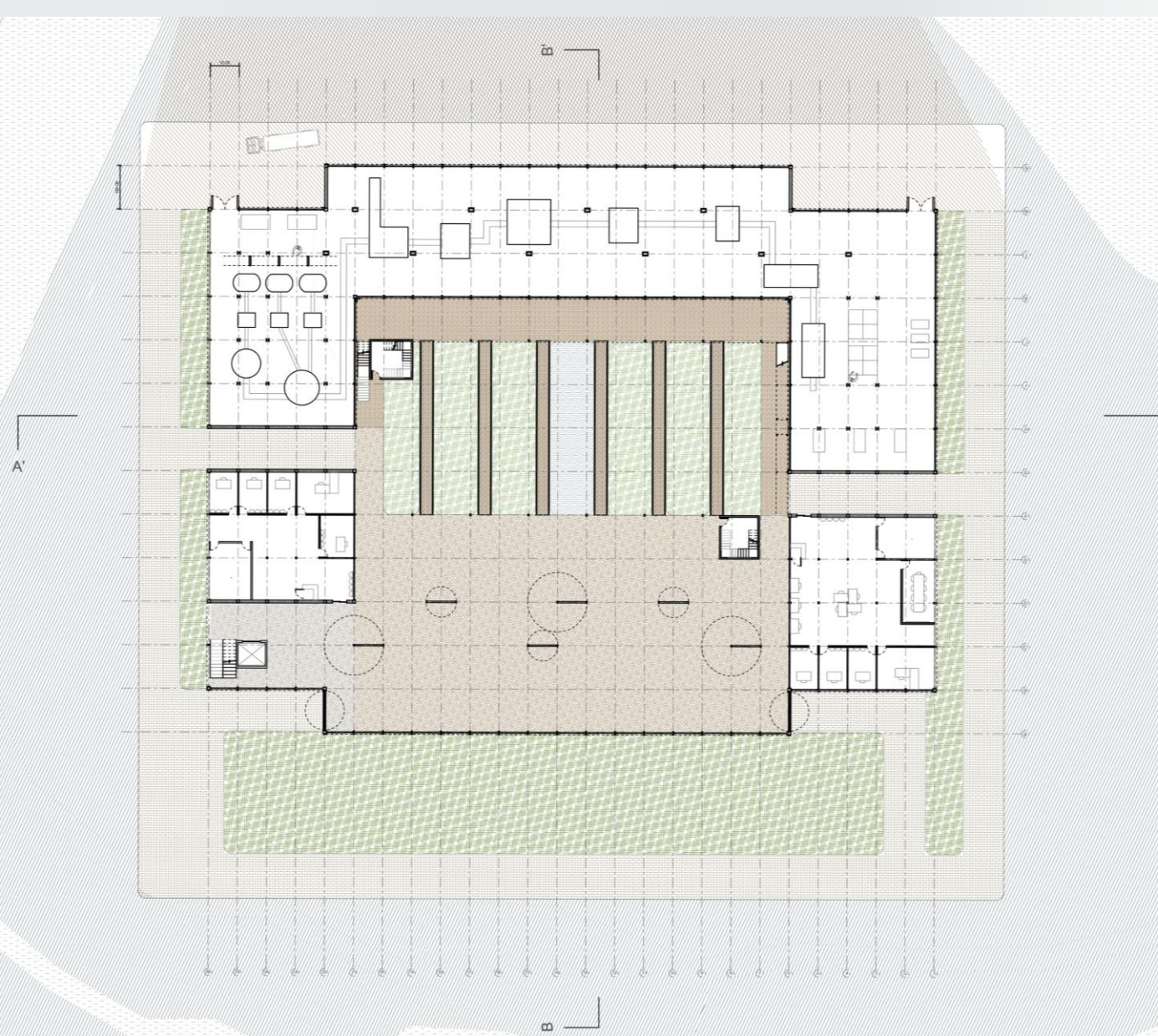
Nestled within the marshy expanse of Bayou Vista, Resource Box is a pioneering fusion of industrial ingenuity and ecological restoration. Designed as a dual-function facility, it seamlessly intertwines two critical missions: recycling post-consumer plastic and restoring wetland ecosystems. This innovative approach highlights the building's commitment to sustainable resource management and environmental renewal. Public engagement is a cornerstone of the design. The first floor houses the industrial machines that process plastic into yarn, with glass windows that let the public observe the process, while the second floor serves as an observation deck and communal gathering space, offering sweeping views of the surrounding marshlands and promoting ecological awareness through wetland exhibits. Functionally, the building is divided into operational zones to optimize efficiency. The recycling area houses advanced waste-processing systems, while the restoration gallery supports research and educational initiatives to rehabilitate local wetlands. In its entirety, Resource Box is more than a facility; it is a testament to sustainable design.



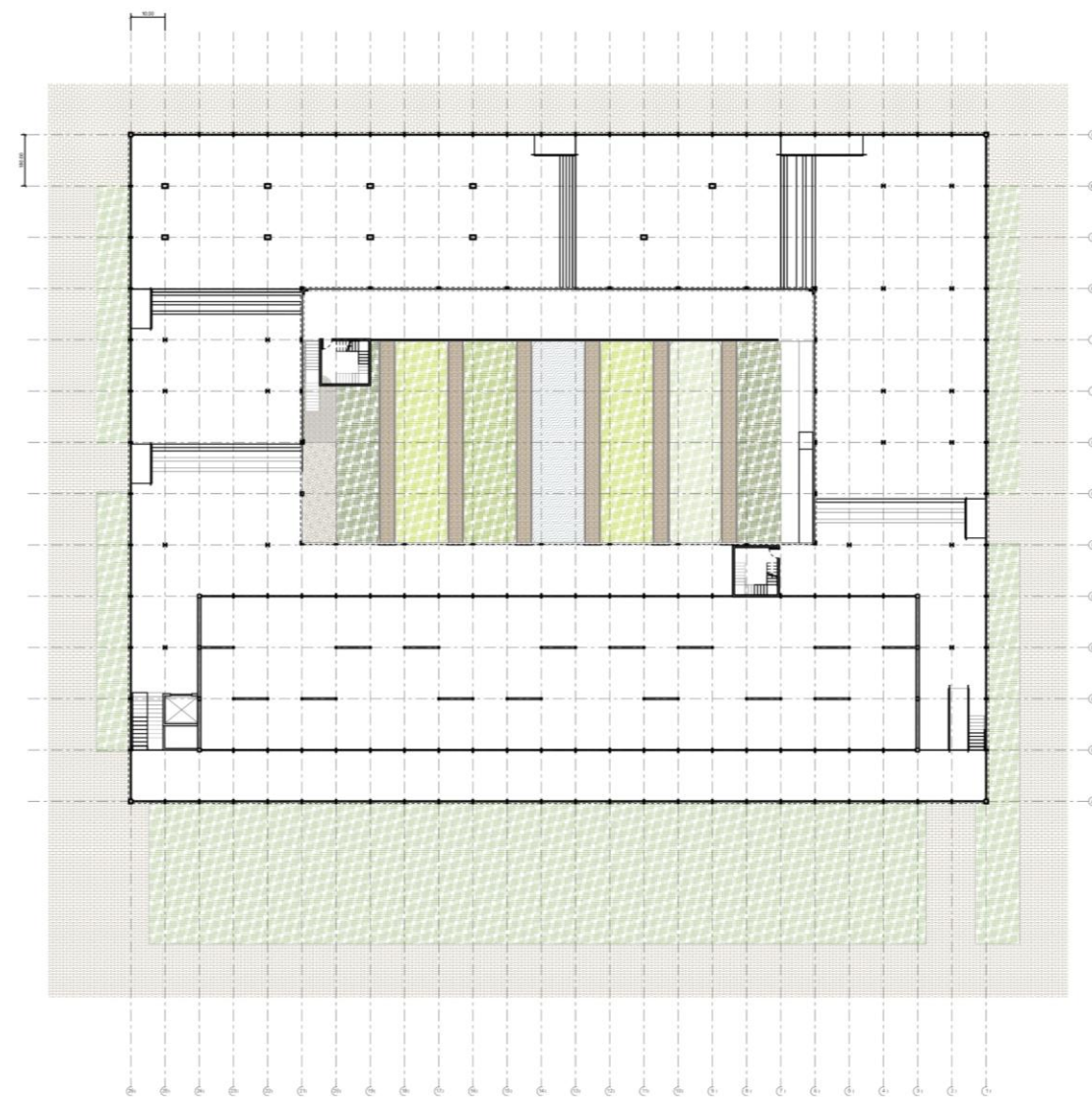
The visitor circuit offers an immersive experience, tracing the journey from waste to renewal. Visitors witness the lifecycle of recycled plastics being transformed into usable yarn.

The facade is defined by expanded metal mesh panels, creating a visually striking and functional envelope. These panels provide essential sun shading, visibility control, and ventilation while reflecting the modularity and transformation inherent in recycling processes. The material's perforated texture allows the building to interact dynamically with light and wind, reinforcing the facility's sustainable ethos.

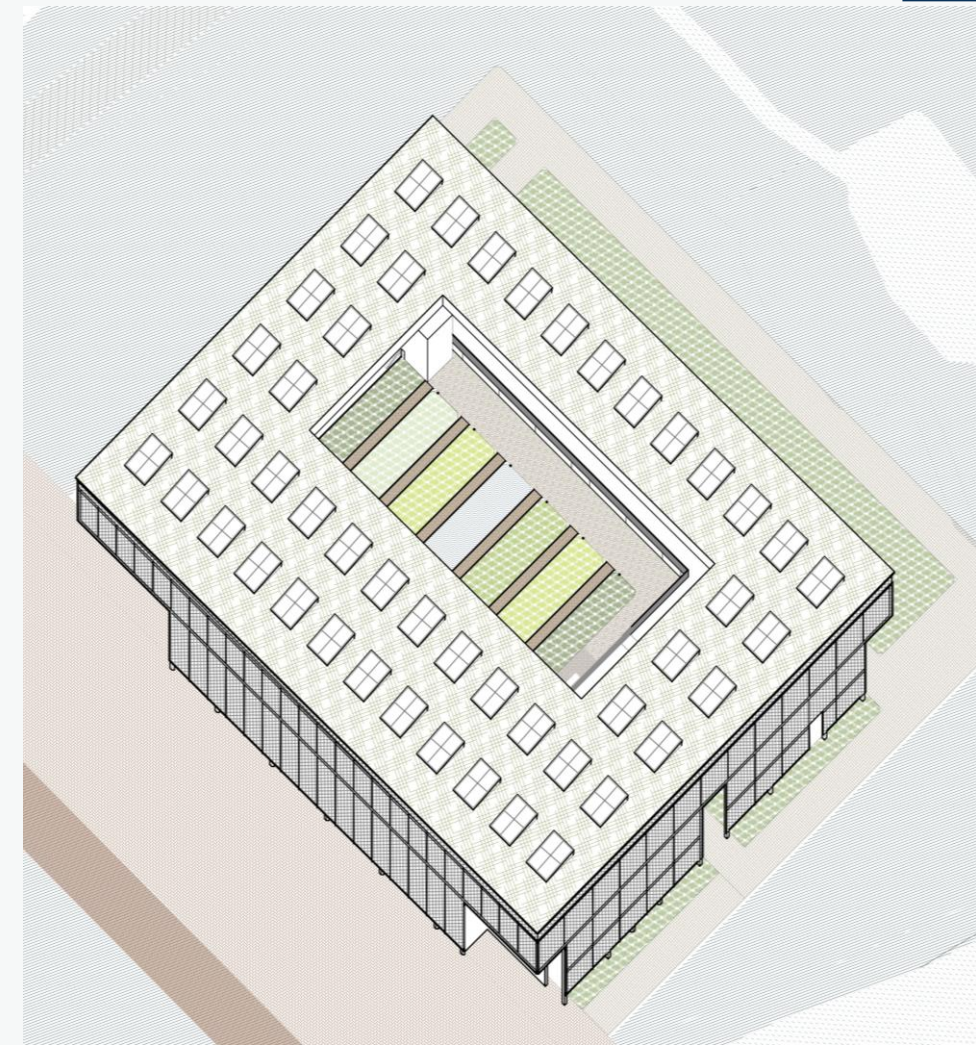
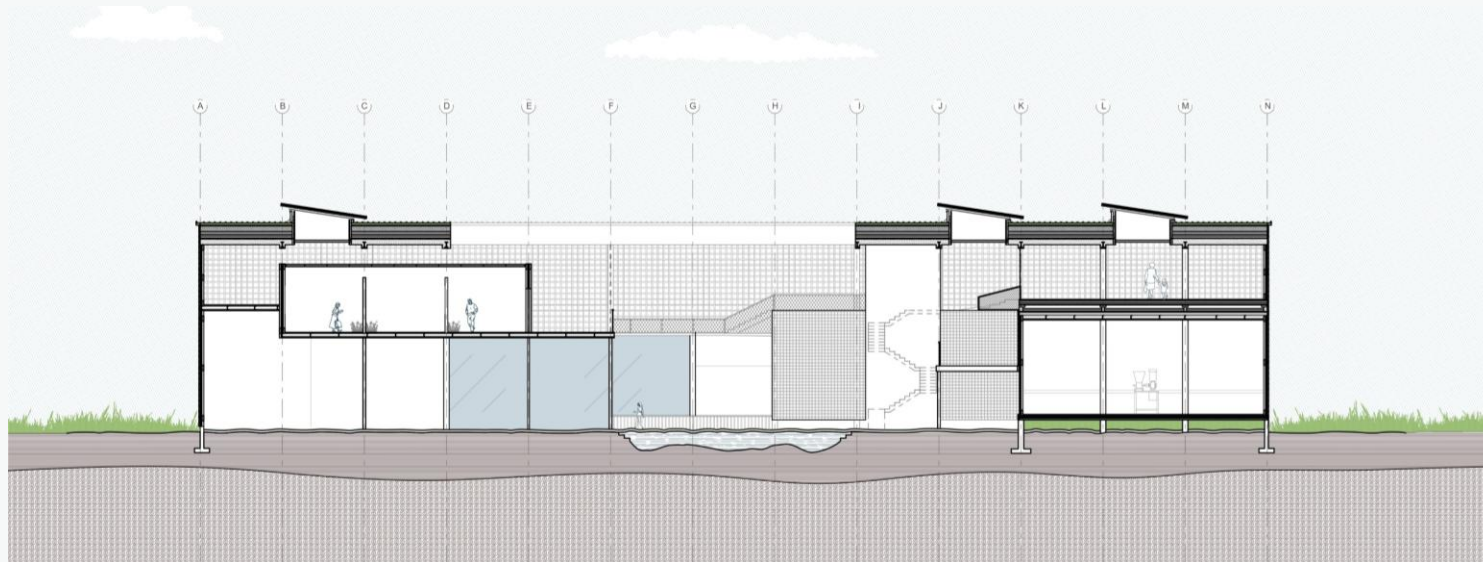




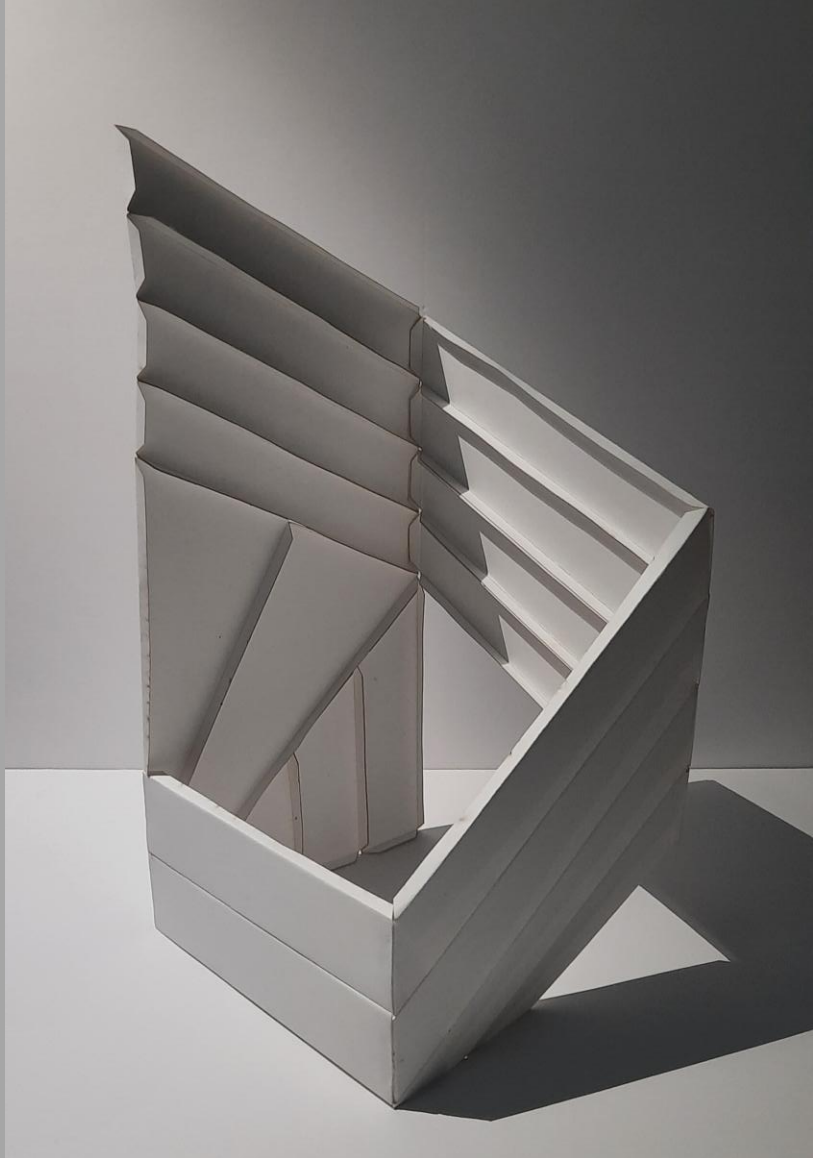
The programmatic layout is rooted in a binary program, comprising two distinct yet interconnected zones: the Plastic Recycling Wing, which addresses the man-made, and the Wetland Restoration Gallery, focused on natural renewal.



These zones converge at the heart of the facility—a shared central courtyard that symbolizes the union of human innovation and ecological stewardship



The large central courtyard, visible from multiple vantage points, serves as a green oasis that bridges industrial activity with natural tranquility in this vast green site. The visitor circuit offers an immersive experience, tracing the journey from waste to renewal.



04. MISCELLANEOUS

2022 - 2025

Displaying concept models and photographs taken .







THANK YOU

Meghana Kankaria
Student, University of Houston

